

AjJay Kannan

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Summary

Data scientist who ships production AI across the full stack spanning classical ML, mathematical optimization, and LLM systems, for supply chain and enterprise operations. Over 3+ years at CHS: built a substitute-product recommender using LLM attribute extraction and embedding similarity that lifted in-full-on-time delivery by 16%, an LLM-agent code-review system that cut issue-detection time by 95%, and end-to-end data platform and workflow automation initiatives that eliminated 1,000+ manual hours/month. Own the work end-to-end — from framing the workflow to deploying tools teams trust and adopt.

Skills

Programming Python, SQL, R

Platform AWS, Azure, Snowflake, dbt, Dataiku, MongoDB

DevOps/Infra Docker, Git, CI/CD with Azure Pipelines

Apps/BI Streamlit, Dash, Flask, Bootstrap, Power BI

ML/AI TensorFlow, Pytorch, Prophet, Optimization

LLM Langchain, Llamaindex, Snowflake Cortex

Experience

CHS Inc

Data Scientist — Supply Chain Planning and Execution

Previously: Data Analyst Intern

St. Paul, MN

Jun 2023 – Present

Jan 2023 – Apr 2023

- Led **end-to-end design, architecture, and deployment** of **scalable data and AI platforms** for supply chain and operations, partnering with cross-functional teams to drive **production adoption and change management**
- Engineered an **LLM-driven documentation workflow** for data products, cutting manual effort by **~40 hours per week**
- Developed an **automated code-review system** using **LLM agents** that reduced issue detection time by **95%**
- Designed and deployed a **substitute-product recommendation system** through extracting key attributes from large **product-label PDFs** using LLM and using **embedding similarity**; improved search efficiency by **85%** and increased SIFOT by **16%**
- Built scalable **Snowflake/dbt data products** for inventory and terminal operations, enabling **60-day stockout visibility** and **capacity-utilization KPIs** across **50+ terminals**. Eliminated **34 documents** and saved **~750 hours/month**
- Built and supported **demand forecasting and planning application** using statistical and ML models (**Prophet, LSTM**), reducing planning cycle time by **20%** and improving forecast accuracy by **9%**
- Built a **mixed-integer optimization system** for inventory assignment and blending under **quality and quantity constraints**. Enabled **order-feasibility decisions** and eliminated manual blending workflows, saving **30 hours/week**

Texas A&M University

Data Analyst — Student Technician (Part Time)

College Station, TX

Sep 2021 – Jan 2023

- Reduced churn from **35% to 20%** in the Engineering Honors Program by conducting **Cohort Analysis** in Tableau, applying **K-Means Clustering** for Segmentation, and performing **Hypothesis Testing**

o9 Solutions Inc

Business Operations Specialist Intern

Dallas, TX

Jun 2022 – Aug 2022

- Developed **Python Plugins** for data integration using ETL processes in **Apache Spark** and optimized SQL queries to enhance models for Supply Chain Planning and Control Tower Functionalities

Tata Motors Limited

Operations Manager

Previously: Graduate Engineer Trainee

Pune, India

Aug 2019 – Jun 2021

Aug 2018 – Jul 2019

- Led a 50+ member cross-functional team to deliver 13 CKD vehicle projects, **implementing lean and ML-driven tools** while achieving 100% fulfillment, 23% lower stock-out costs, 4% station downtime, and 30% faster claims closure

Projects

LLM Applications Portfolio

- Developed and deployed multiple LLM prototypes: **LangChain**-based general-AI agent [🔗](#) – **Text-to-SQL agent** [🔗](#) – **RAG using Snowflake Cortex backend** [🔗](#) for interacting with pesticide product label

Microsoft/MarkItDown PDF Plugin [↗](#)

- Built an open-source MarkItDown plugin that converts complex multi-modal PDFs into LLM-ready Markdown for stronger RAG ingestion

Azure Native ReAct RAG Document Summarization Pipeline [↗](#)

- Built an Azure-native ReAct RAG pipeline with **LlamaIndex**, **OpenAI**, **CosmosDB**, and **Azure Functions** for automated document summarization
- Deployed a **Flask web app** with **Bootstrap**, integrated Azure Repos, and implemented **CI/CD** using Azure Pipelines

Quantitative Risk Modeling of Supply Chain Risks on US Automotive GDP [↗](#)

- Built **Bayesian Network** models in AgenaRisk to quantify risks to GDP from trade war, pandemic, and chip shortage

Kaggle Competition - Amex Credit Default Prediction [↗](#)

- Built credit default prediction models using **LightGBM** and **TensorFlow (Neural Decision Forest)**, evaluated with averaged Gini and Recall, achieving scores of 0.7917 (Forest) and 0.7954 (LGBM) vs. a 0.8097 benchmark

Process Parameter Optimization using Design of Experiments and ANOVA

- Statistically optimized forming to achieve a 65° forming angle for SS430 and 0.9 μ surface finish for Inconel

Education

Texas A&M University

Master of Science in Industrial Engineering (GPA: 3.89/4.00)

Aug 2021 – May 2023

PSG College of Technology

Bachelor of Engineering in Production Engineering (GPA: 8.61/10.00)

Jun 2014 – June 2018